

# Solving an Open Problem in Theoretical Physics using AI Assisted Discovery

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## Abstract

This paper demonstrates that artificial intelligence can accelerate mathematical discovery by autonomously solving an open problem in theoretical physics. We present a neuro-symbolic system, combining the Gemini Deep Think large language model with a systematic Tree Search (TS) framework and automated numerical feedback, that successfully derived novel, exact analytical solutions for the power spectrum of gravitational radiation emitted by cosmic strings. Specifically, the agent evaluated the core integral  $I(N, \theta)$  for arbitrary loop geometries, directly improving upon recent AI-assisted attempts [BCE+25] that only yielded partial asymptotic solutions. To substantiate our methodological claims regarding AI-accelerated discovery and to ensure transparency, we detail system prompts, search constraints, and intermittent feedback loops that guided the model. The agent identified a suite of 6 different analytical methods, the most elegant of which expands the kernel in Gegenbauer polynomials  $C_l^{(3/2)}$  to naturally absorb the integrand's singularities. The methods lead to an asymptotic result for  $I(N, \theta)$  at large  $N$  that both agrees with numerical results and also connects to the continuous Feynman parameterization of Quantum Field Theory. We detail both the algorithmic methodology that enabled this discovery and the resulting mathematical derivations.

## 1 Introduction

A central challenge in evaluating the reasoning capabilities of modern Large Language Models (LLMs) is demonstrating their ability to solve novel, open mathematical problems. In this paper, we demonstrate the power of AI-accelerated mathematical discovery by deploying a neuro-symbolic system to solve a challenging problem from theoretical physics: calculating the power spectrum of gravitational radiation emitted by cosmic strings.

The study of cosmic strings as sources of gravitational radiation has seen renewed interest following observations of the stochastic background by Pulsar Timing Arrays. A critical quantity in these studies is the power spectrum  $P_N$  of the  $N$ -th harmonic emitted by a loop. As discussed in Section III.2 of [BCE+25], the power emitted by a Garfinkle-Vachaspati string is governed by the integral:

$$P_N = \frac{32G\mu^2}{\pi^3 N^2} I(N, \theta) \quad (1)$$

where the core integral  $I(N, \theta)$  is defined over the sphere  $\Omega$ :

$$I(N, \theta) = \int d\Omega \frac{[1 - (\cos \theta)^N \cos(N\pi e_1)][1 - (\cos \theta)^N \cos(N\pi e_2)]}{(1 - e_1^2)(1 - e_2^2)} \quad (2)$$

Here,  $e_1 = \hat{r} \cdot \hat{a}$  and  $e_2 = \hat{r} \cdot \hat{b}$  are projection factors depending on the loop opening angle  $\theta$ .

The integrand features singularities at  $e_{1,2} = \pm 1$ , making standard numerical integration unstable and analytical expansion in Legendre polynomials  $P_l(x)$  difficult due to the non-matching weight functions. Previous attempts yielded asymptotic results for large  $N$  or partial solutions for odd  $N$ , but a unified exact solution remained an open problem.

While the mathematical resolution of this integral is interesting in its own right, the primary contribution of this paper is methodological. We provide a rigorous case study in AI-accelerated mathematical discovery, detailing how an LLM reasoning engine, constrained and guided by a Tree Search and automated numerical feedback, can derive novel, closed-form mathematical physics results that have eluded human researchers.

## 2 Methodology: AI-Accelerated Discovery

To solve this problem, we deployed a hybrid neuro-symbolic system combining a large language model reasoning core with a systematic search algorithm.

### 2.1 Gemini Deep Think

The underlying reasoning engine for this experiment was **Gemini** [Deep Think]. This model was tasked with generating mathematical hypotheses, performing symbolic manipulation, and evaluating the “elegance” or viability of proposed derivation steps. Unlike standard prompting, the model was instructed to engage in deep reasoning chains to anticipate downstream convergence issues in infinite series expansions.

### 2.2 Tree Search (TS)

The generation of the proof was managed by a **Tree Search (TS)** algorithm, adapting the code mutation and search strategies described in recent literature[Ayg25].

**State Space:** The search space explored potential basis expansions (e.g., power series, Legendre, Chebyshev, Jacobi, Gegenbauer), using different integration techniques (e.g., contour integration, integration by parts). Each node in the search tree represented a proposed intermediate mathematical expression formatted in L<sup>A</sup>T<sub>E</sub>X, coupled with an automatically generated executable Python function to numerically evaluate it.

**Scoring Metric:** Each node was scored to determine whether the symbolic result agreed with a high-precision numerical calculation of the integral at random parameter values.

**Exploration:** The TS algorithm utilized a predictor plus upper confidence bound (PUCT) approach to balance exploitation and exploration of novel solution strategies. Over the course of the experiment, the system explored approximately 600 unique candidate nodes. The automated Python verifier successfully caught and pruned over 80% of these branches due to algebraic errors or numerical divergence (such as catastrophic cancellation, unstable monomial sums or ill conditioned basis transformations), significantly streamlining the discovery process. Once the algorithm found valid solution paths, we used negative prompting (detailed in Appendix A) to force the discovery of different types of solutions, leading to the six reported here. An example of the sophisticated algorithmic Python code autonomously generated by the model under these negative constraints is provided in Appendix B.

**Tree Structure** The algorithm creates a large tree exploring solutions to the problem. Examples of visualization of trees produced by the algorithm are shown in this link for the examples outlined in the original paper [Ayg25]

### 2.3 Prompts and Intermittent Feedback

To evaluate the capability of the AI to conduct mathematical discovery, we initialized the reasoning engine with a specific system prompt defining the problem constraints within a computational notebook environment (detailed in Appendix A).

Crucially, the Tree Search framework incorporated an automated intermittent feedback loop. When the model proposed an intermediate analytical step, the system evaluated the executable Python code against a high-precision numerical baseline. If the proposed expression exhibited numerical instability, divergence, or an execution error, the evaluation harness caught the exception and injected the traceback and absolute error penalty back into the model’s context window. This automated verification and feedback mechanism allowed the agent to autonomously correct algebraic errors, prune divergent search paths, and converge on exact, mathematically sound derivations.

## 2.4 Methodological Transparency

To fully support our methodological claims and facilitate independent replication, we detail system prompts, the automated evaluation harness, and the negative-prompting constraints utilized during the Tree Search in Appendix A.

The system successfully navigated the space of solutions and managed to identify six different methods. The most elegant and well conditioned derivation uses the Gegenbauer expansion.

## 3 General Structure

The derivations all require recasting the integral into the following form over the unit sphere  $S^2$ :

$$I(N, \mathbf{z}) = \int_{S^2} d\Omega(\mathbf{u}) f_N(\mathbf{u} \cdot \mathbf{z}) f_N(\mathbf{u} \cdot \mathbf{a}) \quad (3)$$

where  $\mathbf{z}$  and  $\mathbf{a}$  are unit vectors with  $\mathbf{z} \cdot \mathbf{a} = \cos t$ . The function  $f_N(t)$  is defined as:

$$f_N(t) = \frac{1 - (-1)^N \cos(N\pi t)}{1 - t^2}, \quad t \in [-1, 1]. \quad (4)$$

For convergence at the poles  $t = \pm 1$ ,  $N$  must be an integer. We define  $A = N\pi$ .

With this general framing, Gemini found six different solutions. In what follows we organize the approaches in three classes: monomial basis approaches, which expand  $f_N(t)$  in the basis  $\{t^{2k}\}$ , spectral basis approaches, which exploit the Funke-Hecke convolution theorem, and an exact solution.

## 4 Class I: Monomial Basis Approaches

### 4.1 Preliminaries: The Taylor Coefficients

We start by expanding  $f_N(t) = \sum_{k=0}^{\infty} d_{2k} t^{2k}$ . The coefficients are found by considering:

$$(1 - t^2) \sum_{k=0}^{\infty} d_{2k} t^{2k} = 1 - (-1)^N \cos(At) = 1 - (-1)^N \sum_{m=0}^{\infty} \frac{(-1)^m (At)^{2m}}{(2m)!}, \quad (5)$$

where we expand the right hand side (RHS) using the cosine power series. Matching coefficients of  $t^{2k}$  yields the set of equations

$$d_{2k} - d_{2k-2} = (-1)^N \frac{(-1)^k A^{2k}}{(2k)!}, \quad \text{with } d_{-2} = 0, \quad (6)$$

implying

$$d_{2k} = (-1)^N \sum_{j=1}^k \frac{(-1)^j A^{2j}}{(2j)!} + (1 - (-1)^N). \quad (7)$$

It is worth noting that this summation involves subtracting large numbers with potential numerical instability when  $A \gg 1$ .

Substituting the expansion into the integral, we obtain:

$$I(N, \mathbf{z}) = \sum_{k=0}^{\infty} \sum_{j=0}^{\infty} d_{2k} d_{2j} J_{2k,2j}(\mathbf{z}) \quad (8)$$

where

$$J_{k,l}(\mathbf{z}) = \int_{S^2} (\mathbf{u} \cdot \mathbf{z})^k (\mathbf{u} \cdot \mathbf{a})^l d\Omega.$$

To complete the derivation we must compute  $J_{k,l}$ , and we now discuss three methods for doing so.

## 4.2 Method 1: Generating Function Approach

We seek  $J_{2k,2j}$  by defining the generating function

$$G(\lambda, \mu) = \int_{S^2} e^{\mathbf{u} \cdot \mathbf{z} + \mu \mathbf{u} \cdot \mathbf{a}} d\Omega, \quad (9)$$

where

$$J_{2k,2j} = \left[ \frac{\partial^{2k}}{\partial \lambda^{2k}} \frac{\partial^{2j}}{\partial \mu^{2j}} G(\lambda, \mu) \right]_{\lambda=\mu=0}. \quad (10)$$

Letting  $\mathbf{K} = \lambda \mathbf{z} + \mu \mathbf{a}$ , and aligning the polar axis with  $\mathbf{K}$ , we obtain

$$G(\lambda, \mu) = \int_0^{2\pi} d\phi \int_{-1}^1 dt e^{Kt} = 2\pi \left[ \frac{e^{Kt}}{K} \right]_{-1}^1 \quad (11)$$

$$= 2\pi \frac{e^K - e^{-K}}{K} = 4\pi \frac{\sinh K}{K}, \quad (12)$$

where  $K = |\mathbf{K}|$ , with  $K^2 = (\lambda \mathbf{z} + \mu \mathbf{a}) \cdot (\lambda \mathbf{z} + \mu \mathbf{a}) = \lambda^2 + \mu^2 + 2\lambda\mu \cos \theta$ . This then implies

$$\frac{\sinh K}{K} = \sum_{s=0}^{\infty} \frac{K^{2s}}{(2s+1)!} = \sum_{s=0}^{\infty} \frac{(\lambda^2 + \mu^2 + 2\lambda\mu \cos \theta)^s}{(2s+1)!}. \quad (13)$$

Given this result we can directly compute  $J_{2k,2j}$  using Eqn. 10, resulting in an explicit sum involving factorials and powers of  $\cos \theta$ .

## 4.3 Method 2: Gaussian Integral Lifting

Here we evaluate  $J_{2k,2j}$  by lifting the basic equation to a Gaussian Integral, via

$$M = \int_{\mathbb{R}^3} e^{-r^2} (\mathbf{r} \cdot \mathbf{z})^{2k} (\mathbf{r} \cdot \mathbf{a})^{2j} d^3 \mathbf{r}.$$

If we then let  $\mathbf{r} = r \mathbf{u}$ , we obtain:

$$M = \int_0^{\infty} r^2 dr e^{-r^2} r^{2k} r^{2j} \int_{S^2} d\Omega (\mathbf{u} \cdot \mathbf{z})^{2k} (\mathbf{u} \cdot \mathbf{a})^{2j} \quad (14)$$

$$= \left( \int_0^{\infty} r^{2(k+j)+2} e^{-r^2} dr \right) J_{2k,2j}. \quad (15)$$

The radial integral can be evaluated directly as  $\frac{1}{2} (k+j+3/2)$ , implying

$$M = \frac{1}{2} \left( k+j+\frac{3}{2} \right) J_{2k,2j}. \quad (16)$$

At the same time, we can use the identity

$$(\mathbf{r} \cdot \mathbf{z})^{2k} = \partial^{2k} |_{\mathbf{r}=0} e^{-\mathbf{r} \cdot \mathbf{z}}$$

to find

$$M = \partial_\mu^{2k} \partial_\mu^{2j} \Big|_{\mu=0} \int_3 e^{-r^2 + \mathbf{r} \cdot (\mathbf{z} + \mu \mathbf{a})} d^3 \mathbf{r}. \quad (17)$$

The integral can be evaluated analytically since it is a standard Gaussian Integral. Namely, we have that

$$\int e^{-r^2 + \mathbf{b} \cdot \mathbf{r}} = \pi^{3/2} e^{b^2/4},$$

where  $\mathbf{b} = \lambda \mathbf{z} + \mu \mathbf{a}$ , so  $b^2 = \lambda^2 + \mu^2 + 2\lambda\mu \cos \theta$ . This implies the analytical formula for  $M$

$$M = \pi^{3/2} \partial_\mu^{2k} \partial_\mu^{2j} \Big|_{\mu=0} \exp \left( \frac{\lambda^2 + \mu^2 + 2\lambda\mu \cos \theta}{4} \right). \quad (18)$$

Expanding the exponential and equating (16) yields the value of  $J_{2k,2j}$ .

#### 4.4 Method 3: Hybrid Coordinate Transformation

The next method proceeds using a different approach, projecting the power series onto a Legendre basis as follows. We expand

$$t^{2k} = \sum_{m=0}^k \mathcal{T}_{k,m} P_{2m}(t), \quad (19)$$

where the coefficients  $\mathcal{T}_{k,m}$  are known analytically involving factorials.

We then substitute this formula into the Taylor expansion

$$f_N(t) = \sum_{k=0}^{\infty} d_{2k} \left( \sum_{m=0}^k \mathcal{T}_{k,m} P_{2m}(t) \right) = \sum_{m=0}^{\infty} \left( \sum_{k=m}^{\infty} d_{2k} \mathcal{T}_{k,m} \right) P_{2m}(t), \quad (20)$$

and define  $C_{2m} = \sum_{k=m}^{\infty} d_{2k} \mathcal{T}_{k,m}$ , and calculate the integral using the spectral form we derive in the next section (Eq. 21). Note that this derivation is more unstable than the spectral basis approaches defined below, since the calculation of  $C_{2m}$  relies on the unstable  $d_{2k}$ .

## 5 Class II: Spectral Basis Approaches

We now turn to methods that are based entirely on spectral bases, exploiting the Funk-Hecke Convolution Theorem. Since  $I(N, \cdot)$  is a spherical convolution, if  $f_N(t) = \sum C_{2j} P_{2j}(t)$ , then we can use the Funk-Hecke Theorem to show the following:

$$I(N, \cdot) = 4\pi \sum_{j=0}^{\infty} \frac{C_{2j}^2}{4j+1} P_{2j}(\cos \theta). \quad (21)$$

The problem then reduces to finding  $C_{2j}$ . In what follows we present two methods for doing this without using the Taylor series.

### 5.1 Method 4: Spectral Galerkin (Matrix Method)

We define the problem as a linear system in the Legendre basis. Let us define

$$g(t) \equiv (1 - t^2) f_N(t) = 1 - (\cos \theta)^N \cos(At).$$

If we then substitute  $f_N = \sum_j C_{2j} P_{2j}$  and project onto test function  $P_{2i}$ , we obtain

$$\sum_j C_{2j} \int_{-1}^1 (1 - t^2) P_{2i}(t) P_{2j}(t) dt = \int_{-1}^1 P_{2i}(t) g(t) dt. \quad (22)$$

This equation is of the form  $\mathbf{G}\mathbf{C} = \mathbf{b}$ . Let  $G_{ij}$  be the matrix element of  $\mathbf{G}$ , we can use the identity  $t^2 P_l = A_l P_{l+2} + B_l P_l + C_l P_{l-2}$ , to obtain an explicit formula

$$G_{ij} = \int_{-1}^1 P_{2i} P_{2j} dt - \int_{-1}^1 P_{2i} (t^2 P_{2j}) dt \quad (23)$$

$$= \frac{2\delta_{ij}}{4i+1} - \int_{-1}^1 P_{2i} (A_{2j} P_{2j+2} + B_{2j} P_{2j} + C_{2j} P_{2j-2}) dt. \quad (24)$$

Orthogonality then implies that the integral is non-zero only when  $2i = 2j + 2, 2j, 2j - 2$ . Thus,  $G$  is a symmetric positive definite tridiagonal matrix.

To construct the right hand side,  $b_i = \int P_{2i} [1 - (-1)^N \cos(At)] dt$ , we use the Bauer plane wave expansion

$$\cos(At) = \sum_{l=0}^{\infty} (-1)^l (4l+1) j_{2l}(A) P_{2l}(t). \quad (25)$$

The orthogonality of  $P_{2i}$  then implies

$$b_i = \frac{2\delta_{i0}}{1} - (-1)^N (-1)^i (4i+1) j_{2i}(A) \frac{2}{4i+1} = 2\delta_{i0} - 2(-1)^{N+i} j_{2i}(A). \quad (26)$$

Given that  $G$  is tridiagonal and positive definite, we can solve  $\mathbf{G}\mathbf{C} = \mathbf{b}$  to find the coefficients  $C_{2j}$  in  $O(N)$  time.

## 5.2 Method 5: Spectral Volterra (Recurrence Method)

Here we derive a forward recurrence for  $C_{2j}$ . We start with

$$C_l = \frac{2l+1}{2} \int_{-1}^1 f_N(t) P_l(t) dt = \frac{2l+1}{2} \gamma_l,$$

and then derive a recurrence relation for  $\gamma_l$  by multiplying the Legendre differential equation by  $f_N(t)$  and integrating, implying

$$l(l+1)\gamma_l = \int_{-1}^1 f_N(t) \frac{d}{dt} [(1-t^2)P'_l(t)] dt. \quad (27)$$

Integration by parts then implies

$$l(l+1)\gamma_l = \int_{-1}^1 f'_N(t) (1-t^2) P'_l(t) dt. \quad (28)$$

Let  $H(t) = (1-t^2)f_N(t) = 1 - (-1)^N \cos(At)$ . Differentiating this product gives:

$$(1-t^2)f'_N(t) = H'(t) + 2tf_N(t). \quad (29)$$

Substituting this back into the integral, we split the result into two terms,  $T_1(l)$  and  $T_2(l)$ :

$$l(l+1)\gamma_l = \underbrace{\int_{-1}^1 H'(t) P'_l(t) dt}_{T_1(l)} + \underbrace{\int_{-1}^1 2tf_N(t) P'_l(t) dt}_{T_2(l)}. \quad (30)$$

We first evaluate  $T_1(l)$ . We have  $H'(t) = \frac{d}{dt} [1 - (-1)^N \cos(At)] = (-1)^N A \sin(At)$ .

$$T_1(l) = \int_{-1}^1 (-1)^N A \sin(At) P'_l(t) dt. \quad (31)$$

Integrating by parts again:

$$T_1(l) = [(-1)^N A \sin(At) P_l(t)]_{-1}^1 - \int_{-1}^1 (-1)^N A^2 \cos(At) P_l(t) dt. \quad (32)$$

For integer  $N$ ,  $\sin(A) = \sin(N\pi) = 0$ , so the boundary term vanishes.

$$T_1(l) = (-1)^N A^2 \int_1^1 \cos(At) P_l(t) dt. \quad (33)$$

Using the plane wave expansion identity  $\int_1^1 e^{iAt} P_l(t) dt = 2i^l j_l(A)$  and taking the real part for even  $l = 2j$ :

$$T_1(2j) = -2A^2 (-1)^{N+j} j_{2j}(A), \quad (34)$$

where  $j_{2j}(A)$  is the spherical Bessel function of the first kind.

We now turn to the term  $T_2(l)$  by expanding  $2tP'_l(t)$  in the Legendre basis. Using standard identities and rearranging terms, we arrive at a telescoping sum structure:

$$T_2(2j) = 2 \sum_{m=0}^{j-1} [(2m+2)\gamma_{2m+2} + (2m+1)\gamma_{2m}]. \quad (35)$$

This implies

$$T_2(2j) = 4j\gamma_{2j} + S_R(j), \quad (36)$$

where  $S_R(j)$  is a running sum of lower-order coefficients:

$$S_R(j) = \sum_{m=0}^{j-1} (8m+2)\gamma_{2m}. \quad (37)$$

Substituting  $T_1$  and  $T_2$  back into the main equation then implies

$$2j(2j+1)\gamma_{2j} = T_1(2j) + 4j\gamma_{2j} + S_R(j), \quad (38)$$

or:

$$\frac{4(2j^2 - j)}{4j+1} C_{2j} = T_1(2j) + S_R(j). \quad (39)$$

We can then compute  $C_{2j}$  sequentially from  $C_0$  in  $O(N)$  time.

## 6 Class III: The Analytic Solution

### 6.1 Method 6: The Gegenbauer Method

The final method identifies the exact analytic solution for the coefficients  $C_{2j}$  by choosing a basis orthogonal with respect to the weight  $w(t) = 1 - t^2$ , which naturally cancels the singularity in  $f_N(t)$ .

The first step is to expand  $f_N(t)$  in the basis of Gegenbauer polynomials  $C_l^{(3/2)}(t)$ ,

$$f_N(t) = \sum_{m=0}^{\infty} b_{2m} C_{2m}^{(3/2)}(t), \quad (40)$$

and determine the coefficients by orthogonality:

$$b_{2m} = \frac{1}{h_{2m}} \int_1^1 \underbrace{\left[ \frac{1 - (-1)^N \cos(At)}{1 - t^2} \right]}_{f(t)} C_{2m}^{(3/2)}(t) \underbrace{(1 - t^2)}_{\text{weight}} dt, \quad (41)$$

where  $h_{2m} = \frac{2(2m+1)(2m+2)}{4m+3}$  is the standard normalization constant for the Gegenbauer polynomials  $C_{2m}^{(3/2)}(t)$ .

Crucially, the weight cancels the denominator, leaving a regular integral of the numerator  $G_N(t) = \frac{1 - (-1)^N \cos(At)}{1 - t^2}$ .

We then use the identity  $C_k^{3/2}(t) = P'_{k+1}(t)$ , and integrate by parts. The boundary terms vanish because  $G_N(\pm 1) = 0$ , and the integral reduces to the Fourier transform of Legendre polynomials:

$$b_{2m} = A(-1)^{N+m} j_{2m+1}(A) \frac{4m+3}{(2m+1)(2m+2)}. \quad (42)$$

Using the expansion  $C_{2m}^{3/2}(t) = \sum_{j=0}^m (4j+1)P_{2j}(t)$ , we then relate the Legendre coefficients  $C_{2j}$  to the Gegenbauer coefficients  $b_{2m}$  via a tail sum:

$$C_{2j} = (4j+1) \sum_{m=j}^{\infty} b_{2m}. \quad (43)$$

We can simplify the formula further by noting that if we set  $j = 0$ , the sum of all Gegenbauer coefficients must equal  $C_0$ :

$$C_0 = \sum_{m=0}^{\infty} b_{2m}, \quad (44)$$

so that

$$C_{2j} = (4j+1) \left( C_0 + A(-1)^N \sum_{m=0}^j (-1)^m j_{2m+1}(A) \frac{4m+3}{(2m+1)(2m+2)} \right). \quad (45)$$

We can then find an analytical expression for  $C_0$  by noting

$$C_0 = \frac{2(0)+1}{2} \int_{-1}^1 f_N(t) P_0(t) dt = \frac{1}{2} \int_{-1}^1 \frac{1 - (-1)^N \cos(At)}{1 - t^2} dt, \quad (46)$$

and then exploiting the partial fraction decomposition  $\frac{1}{1-t^2} = \frac{1}{2} \left( \frac{1}{1-t} + \frac{1}{1+t} \right)$ , implying

$$\int_{-1}^1 \frac{1 - (-1)^N \cos(At)}{1 - t^2} dt = \int_0^{2A} \frac{1 - \cos(x)}{x} dx. \quad (47)$$

Using the substitution  $x = A(1-t)$ , we note that  $\cos(At) = \cos(A-x) = (-1)^N \cos(x)$  since  $A = N\pi$ . The resulting integral is the standard definition of the generalized cosine integral function  $\text{Cin}(z)$ :

$$\text{Cin}(z) \equiv \int_0^z \frac{1 - \cos(t)}{t} dt = \gamma + \ln(z) - \text{Ci}(z). \quad (48)$$

Thus, we obtain the exact formula

$$C_0 = \frac{1}{2} \text{Cin}(2A). \quad (49)$$

This provides a closed-form solution for the spectral coefficients without matrix inversion or recurrence.

## 6.2 Hierarchical Verification and Refinement

The initial set of solutions, including the six methods detailed above, was generated by the Tree Search (TS) framework utilizing a standard Gemini Deep Think configuration. This automated exploration identified diverse approaches, notably the Gegenbauer method (Method 6), which initially provided an exact solution expressed as an infinite tail sum of coefficients. While mathematically sound, this result was not yet in a fully closed form.

To achieve the final, fully analytic form, a human researcher manually initiated a new interaction session, prompting a larger, more advanced version of Gemini Deep Think with the intermediate TS results. The model was explicitly tasked to rigorously verify the existing proofs and search for further simplifications. This step represents a synergistic human-AI handoff rather than a fully autonomous pipeline. During this interactive process, the advanced model independently identified and corrected the aforementioned error in the initial formulation of the Spectral Volterra Recurrence (Method 5), where the  $4j+1$  denominator



dependency had been missed. By establishing the equivalence between the corrected Method 5 and Method 6, the model recognized that the localized recurrence structure allowed the infinite tail sum in Method 6 to be telescoped. This insight, derived during the interactive verification phase, led to the finite closed-form expression involving the Complementary Cosine Integral, improving the solution from an exact infinite series to a fully analytical finite form.

## 7 Comparison of full solution with numerical calculations

Finally we present a comparison of the results of different methods with numerical calculations of the integral for different values of  $N$ ,  $\alpha$ . Figure 1 shows a comparison for the exact solution Method 6 over a range of  $N$  and  $\alpha$ . The agreement with the numerical calculation is excellent.

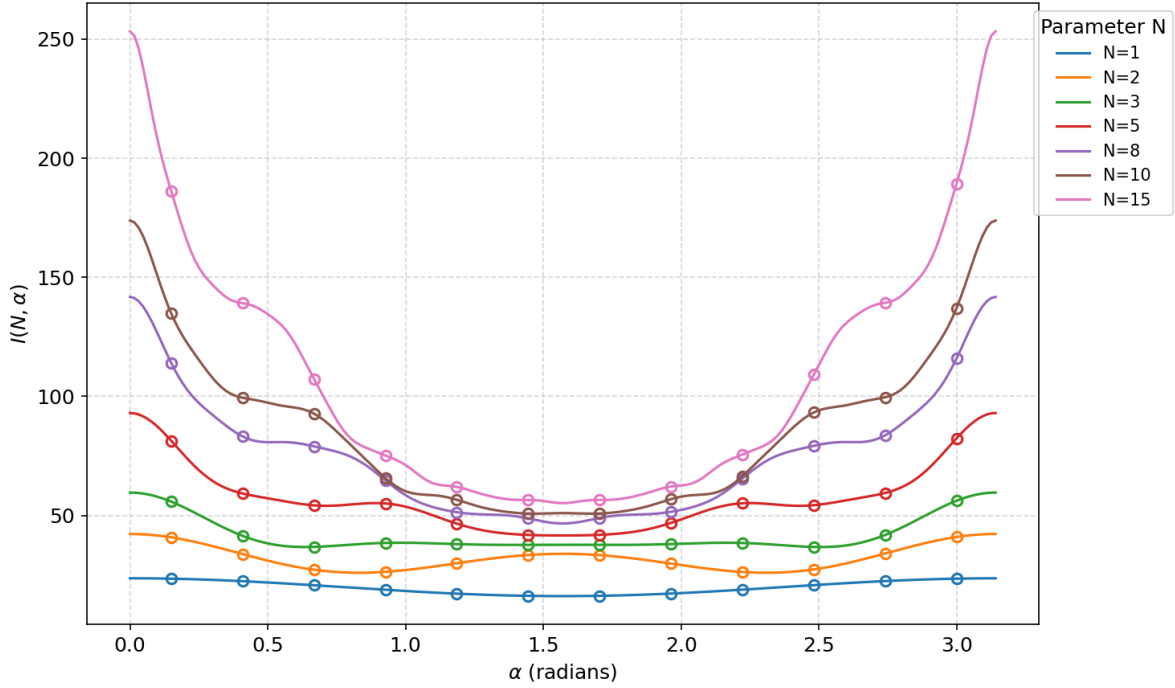


Figure 1: **Verification of the Method 6 analytical solution.** The solid curves represent the closed-form expression derived for the integral  $I(N, \alpha)$ . Open circles represent reference values obtained via direct numerical integration. The excellent agreement across a range of parameters  $N$  and  $\alpha$  validates the derivation. We note that for this range of  $N$  all of the methods achieve similar level of accuracy.

While at small  $N < O(15)$  all methods converge and give similar accuracy, as  $N \rightarrow \infty$  the monomial expansion methods exhibit numerical instability. There are also significant speed tradeoffs of the different methods. Figure 2 compares the methods for  $I(N = 20, \alpha)$ . To clearly separate the stable methods from the diverging monomial methods, the top panel plots the absolute error  $|I_{\text{method}} - I_{\text{numerical}}|$  on a logarithmic scale. The bottom panel compares the execution speed. The second method completely fails at this  $N$  due to numerical issues. We also note a distinct spike in the computation time for Method 5 around  $\alpha \approx 1.05$ ; this is due to transient matrix conditioning issues near specific roots of the Legendre basis at that angle.

## 8 Asymptotics at large

Finally, we turn to a discussion of the asymptotics of  $I(N, \alpha)$  at large  $N$ .

For this we will start with the exact spectral coefficients  $b_{2m}$  derived with methods 4, 5, 6 and take the macroscopic limit  $N \rightarrow \infty$ . We first note that the spherical Bessel function has the leading order

### Comparison of Spherical Integral Calculation Methods (N=20)

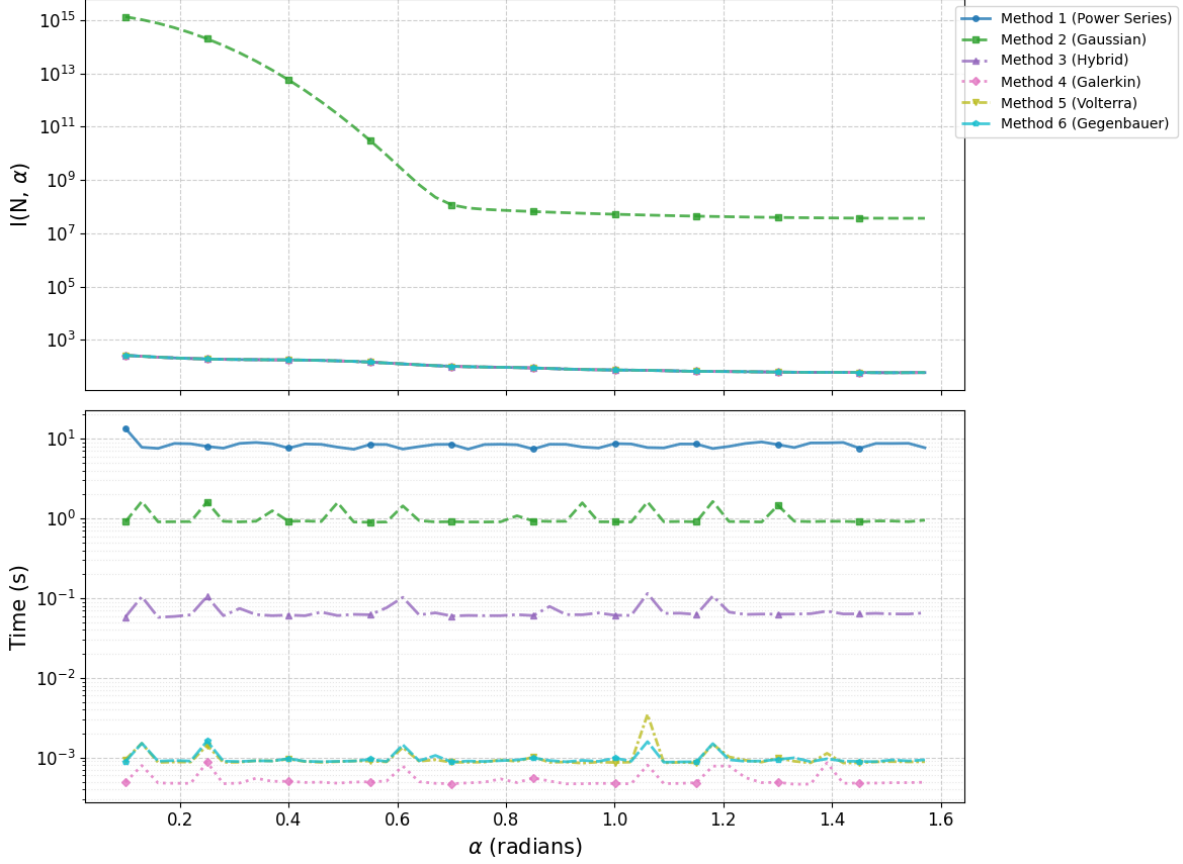


Figure 2: **Comparison of methods: absolute error and speed for  $N = 20$ .** The top panel shows the absolute error  $|I_{\text{method}} - I_{\text{numerical}}|$  of the different methods for  $N = 20$  as a function of  $\alpha$  on a logarithmic scale. The stable spectral methods hug the numerical noise floor, while Method 2 diverges and fails due to numerical instability. The bottom panel compares the speed of the methods, with the spectral methods evaluating orders of magnitude faster. We also note a distinct spike in computation time for Method 5 around  $\alpha \approx 1.05$ ; this corresponds to a transient matrix conditioning issue near a root of the associated Legendre polynomials.